

Data Visualization Lab-02

Duration: 2 Hours

Total Marks: 100

Scenario

You are working as a Business Intelligence Developer for a multi-channel e-commerce company. The company receives sales data from three different platforms: Website, Mobile App, and Marketplace.

The datasets contain inconsistent formats, missing values, duplicates, and different naming conventions. Your task is to clean, standardize, transform, and combine the data into one unified model suitable for reporting.

Lab Objectives

- Students must:
- Clean messy datasets using Power Query
- Standardize column naming conventions
- Handle null values and duplicates
- Perform column transformations
- Append multiple tables into one unified table
- Apply business logic calculations
- Generate meaningful analytical outputs

Part A – Data Cleaning (Power Query)

1. Remove rows where Order Date is null.
2. Remove unnecessary columns provided in datasets.
3. Rename columns to a standard naming format (OrderID, OrderDate, CustomerID, CustomerName, Product, Quantity, UnitPrice, Region, Status, DiscountRate, Channel).

4. Change data types appropriately (Date, Whole Number, Decimal).
5. Remove duplicate records based on OrderID.
6. Remove rows where Customer Name is blank.
7. Replace inconsistent Status values with standardized values (Success, Returned, Cancelled).
8. Convert percentage values (e.g., 5%) into decimal format (0.05).
9. Convert text quantities (e.g., 'three', 'two') into numeric values.
10. Trim and clean text fields to remove extra spaces and hidden characters.

Part B – Column Transformations

1. Split combined columns (e.g., Customer Details) into CustomerID and CustomerName using delimiter.
2. Merge columns to create CustomerKey (CustomerID + '-' + Region).
3. Add a new column called Channel to identify the source of each dataset.
4. Ensure all three datasets follow identical column structure before appending.

Part C – Table Combination

1. Append all three cleaned datasets into a new table named 'All_Sales'.
2. Verify total row count after append.
3. Create measure:

$$\text{Total Net Revenue} = \text{SUMX}(\text{All_Sales}, \text{All_Sales}[\text{Quantity}] * \text{All_Sales}[\text{UnitPrice}] * (1 - \text{All_Sales}[\text{DiscountRate}])))$$
4. Create a measure:

$$\text{Channel Count} = \text{DISTINCTCOUNT}(\text{All_Sales}[\text{Channel}])$$
5. Create a summary of returned orders (count).

$$\text{Returned Orders Count} = \text{CALCULATE}(\text{COUNT}(\text{All_Sales}[\text{OrderID}]), \text{All_Sales}[\text{Status}] = "Returned")$$

Final Deliverables

- Cleaned datasets with proper transformations.

- Unified 'All_Sales' table.
- Data model with correct relationships (if applicable).
- Report page showing:
 1. Revenue by Region
(Hint: Use Clustered Column Chart → Axis: Region → Values: Total Net Revenue → Sort descending.)
 2. Revenue by Channel
(Hint: Use Clustered Bar Chart → Axis: Channel → Values: Total Net Revenue.)
 3. Top 5 Customers by Net Revenue
(Hint: Use Table visual → Add CustomerName + Total Net Revenue → Apply Top N filter (Top 5 by Total Net Revenue).)
 4. Returned Orders Summary
(Hint: Use Card visuals → Returned Orders Count)
 5. Cross-Channel Customer Analysis
(Hint: Use Matrix → Rows: CustomerName → Values: Channel Count)